

Module 8: Cellular Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Describe the stages of the cell cycle (interphase and mitosis)
2. List the phases of mitosis in order (PMAT)
3. Compare mitosis and meiosis
4. Explain how meiosis creates genetic diversity
5. Describe what happens when cell division goes wrong (cancer, nondisjunction)

The Big Picture

Cells divide for two reasons: **Mitosis** = growth and repair (identical copies).

Meiosis = making sex cells/gametes (unique combinations).

Key Terms

Chromosome Basics

- **Chromatin** — Loose DNA (what it looks like between divisions)
- **Chromosome** — Tightly packed DNA (what it looks like during division)
- **Sister Chromatids** — Two identical copies of one chromosome, joined at the centromere
- **Homologous Chromosomes** — Matching pair (one from mom, one from dad); same genes, may have different alleles
- **Diploid (2n)** — Two sets of chromosomes (body cells); humans: $2n = 46$
- **Haploid (n)** — One set of chromosomes (sex cells); humans: $n = 23$

Cell Cycle

- **G₁** — Cell grows

- **S** — DNA is copied (replicated)
- **G₂** — Cell prepares to divide
- **M Phase** — Mitosis + cytokinesis

Mitosis (PMAT)

Phase	What Happens
Prophase	Chromosomes condense; nuclear envelope breaks down
Metaphase	Chromosomes line up in the Middle
Anaphase	Sister chromatids are pulled Apart
Telophase	Two new nuclei form; chromosomes relax
Cytokinesis	Cell splits in two

- Animals: **cleavage furrow** (pinches)
- Plants: **cell plate** (builds a wall)
- Result: **2 identical diploid cells**

Meiosis

- Two rounds of division: Meiosis I and Meiosis II
- **Meiosis I** — Homologs separate (reduction division)
- **Meiosis II** — Sister chromatids separate (like mitosis)
- Result: **4 unique haploid cells**

Mitosis vs. Meiosis

	Mitosis	Meiosis
Divisions	1	2
Daughter cells	2	4
Ploidy	Diploid (2n)	Haploid (n)
Identical to parent?	Yes	No
Purpose	Growth, repair	Make gametes
Crossing over?	No	Yes

Three Sources of Genetic Variation

1. **Crossing over** (Prophase I) — Homologs swap pieces of DNA
2. **Independent assortment** (Metaphase I) — Homologs line up randomly
3. **Random fertilization** — Any sperm + any egg

When Things Go Wrong

- **Nondisjunction** — Chromosomes don't separate correctly → wrong number of chromosomes
- **Aneuploidy** — Having too many or too few chromosomes (e.g., Down syndrome = trisomy 21)
- **Cancer** — Cell cycle checkpoints fail → uncontrolled cell division
- **p53** — A tumor suppressor protein; when it's mutated, cells don't stop dividing

Study Tips

1. **PMAT** — Prophase, Metaphase, Anaphase, Telophase
2. **"Homologs in I, Sisters in II"** — What separates in each meiosis division

3. **Draw it** — Sketch a cell going through mitosis next to one going through meiosis
4. **Think about WHY** — Mitosis = copies, Meiosis = variety
5. **Cancer = broken brakes** — Checkpoints are the brakes; oncogenes are stuck accelerators